

CPU Scheduling Simulator

Test Cases & Expected Results Reference

Scenario A — Basic Mixed Workload

Purpose: Tests normal behaviour with varied arrival times, burst times, and priorities.

Process	Arrival	Burst	Priority
P1	0	8	2
P2	1	4	1
P3	2	9	3
P4	3	5	2
P5	4	2	1

Expected Priority Results

- P1 starts at t=0, preempted at t=1 when P2 arrives (priority 1 > priority 2).
- P2 runs until done, then P5 (also priority 1), then P1 resumes, then P4, then P3.

Expected SRTF Results

- P1 starts at t=0, preempted at t=1 when P2 arrives (BT=4 < remaining=7).
 - P5 preempts when it arrives (BT=2 is shortest remaining time).
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Scenario B — Priority vs Burst Conflict

Purpose: P1 has highest priority but the longest burst — highlights how the two algorithms diverge.

Process	Arrival	Burst	Priority
P1	0	12	1
P2	2	2	3
P3	3	4	3
P4	5	1	4

Algorithm Divergence

Priority Scheduling: P1 runs exclusively until done — it is never preempted because it always holds the highest priority.

SRTF: P2, P3, and P4 each preempt P1 upon arrival because they carry shorter remaining burst times.

Key Observation

- Priority Scheduling is **unfair to P2/P3/P4** — they wait a long time despite being short jobs.
 - SRTF is **unfair to P1** — it keeps getting preempted despite being the most important process.
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Scenario C — Starvation-Sensitive Case

Purpose: Demonstrates starvation. P1 has low priority (3) and long burst (10); P2–P6 arrive continuously with high priority (1).

Process	Arrival	Burst	Priority
P1	0	10	3
P2	1	1	1
P3	2	1	1
P4	3	1	1
P5	4	1	1
P6	5	1	1

Starvation Analysis

- **Priority Scheduling:** P1 is preempted every time a new high-priority process arrives. P1 waits an extremely long time — **classic starvation demonstrated**.
 - **SRTF:** P1 is preempted by short-burst arrivals for a similar starvation effect, but for a different algorithmic reason.
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Scenario D — Validation Case (Invalid Input)

Purpose: Verifies that the simulator catches bad input before attempting simulation and surfaces descriptive error messages.

Process	Arrival	Burst	Priority
(empty)	0	5	1

P2	-1	4	2
P3	0	0	3

Expected Validation Errors

- **Row 1:** Process ID cannot be empty.
 - **Row 2:** Arrival time must be ≥ 0 .
 - **Row 3:** Burst time must be > 0 .
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How to Verify Results Manually

For each process after simulation, compute the following metrics:

CT	Time when process finishes (read from Gantt chart)
TAT	CT – Arrival Time
WT	TAT – Burst Time
RT	First time on CPU – Arrival Time
Avg	Sum of metric / total number of processes

Read CT directly from the Gantt chart. All other values derive from it using the formulas above.